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Project Title:	Resonance Effects in Single- and Double-Diffusive Systems Under Gravity Modulation		
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Key Personnel Changes/Previous PI:	Chen, Chuan F.		
COI Name (Institution):	Chen, Chuan F. (University of Arizona)		
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Task Description:	The plan of work includes linear stability analysis, experiments, and numerical simulation, all carried out concurrently. Linear stability analysis will be performed to identify the resonance effect of gravity modulation and to gain better insight in the observed convection. The experimental investigation consists of two series of experiments, one with a vertical layer and the other with a horizontal double-diffusive layer. The numerical simulation investigations serve two purposes. One is to check the results of linear stability analysis by including the actual geometry of the experimental tank, thereby improving on the preliminary design of the experiment. The second purpose is to provide detailed information on the nonlinear evolution of the flow field in the supercritical state. The objective of this research is to study and to understand the effect of gravity modulation, including the resonance effects, on the stability and subsequent convection in single- and double- diffusive systems. The expected significance of this research is an understanding of the effect of g-jitter on convection in metallic alloy melts during material processing and on induced mixing process.		

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

During the past year, our effort was focused on the horizontal double-diffusive layer.
1 Experiment - Horizontal Tank

The experiments were carried out in a horizontal test tank 50mm wide x 100mm long x 10mm deep. The lower heat transfer wall is made of copper. The upper heat transfer boundary is transparent for the purpose of PIV application. It is made of sapphire 1mm thick, which forms the lower boundary of a transparent Plexiglass box. Transparent glycerin-water solution of constant temperature is circulated through the box to cool the fluid layer below.

The double-diffusive convection experiment was carried out with ethanol-water solution. The lower half of the tank was filled with pure water first, and then the upper half with 2.0 %(wt) ethanol-water solution slowly to minimize mixing. Both fluids were seeded with 10μm polycrystalline particles for PIV purposes. Once the tank is filled, the top cover is secured. Solution is squeezed in from one of the two tubing on top to remove any bubble trapped in the tank. The tank is left standing for 30 minutes to allow diffusion to smooth out the concentration distribution. At this time, a temperature difference is imposed (in this particular experiment ~12oC), with higher temperature at the bottom surface. The thermal diffusion time (1 cm length scale) for 1%(wt) ethanol water solution is about 3 minutes. Consequently, the linear temperature profile is established quickly. Motion in the fluid is detected at 95 minutes after the temperature difference is imposed. Effectively, the thermal Rayleigh number RT =162,000 is fixed, and the stratification is continually weakened by diffusion as time goes on. Eventually the system becomes unstable and motion onsets in the fluid. The actual concentration distribution just prior to the onset of instabilities is obtained by numerically integrating the 1-D diffusion equation. The distribution can be approximated very closely by a sinusoidal curve. The critical solute Rayleigh number RS, based on the calculated concentration difference between the top and the bottom boundaries, is 126,000.

The onset motion was captured by video camera. We can also see clearly the motion started at the lower left part of the tank. There are two layers of re-circulating cells in the first 10 minutes. This is in agreement with the linear theory. Then, the two layers merge to become one. The recirculation becomes faster and faster. This same behavior is predicted by numerical simulation. Spectral analysis of the transient velocity data at a fixed point revealed that the circulation is oscillatory with a dominant frequency of 0.043Hz. This again agrees with the linear theory prediction.

The effect of gravity modulation is investigated by vertically oscillating the test tank at a frequency of 1 Hz with amplitude of 10cm, a maximum additional gravitational acceleration of 0.4G. Onset of convection occurs at 86 min after the temperature difference is imposed at RS = 136,000. The system is slightly less stable than the steady-G case. The velocity field shows a single layer structure most of the time. Two-layer structures are seen sometimes, but they quickly evolve into a single layer. At onset, the dominant frequency is 0.057Hz, a 32% increase from the steady-G case.

2 Linear Stability Analysis

We consider a stably stratified horizontal layer subject to heating from the bottom. The basic state is quiescent fluid with linear temperature (vertical) distribution and sinusoidal concentration (vertical) distribution. The sinusoidal concentration distribution is a good approximation to the experimental condition. The dimensionless parameters relevant for the problem are Pr, Le, RTand RS. We use Galerkin method to perform normal mode analysis. Chandrasekhar functions are used for the streamfunction expansion while sine and cosine functions are used for the temperature and concentration expansions. The effect of gravity modulation is investigated by using Floquet theory. For the experimental case, Pr = 7 and Le = 118.3. We use the linear stability analysis to determine the critical conditions for a given RS = 126,000, the experimental value. Under steady G, the sinusoidal concentration distribution is found to be more stable than the linear distribution. The critical RT is 158,000 and the corresponding frequency is 0.045Hz, both in good agreement with the experiment. By examining the dispersion curve of the onset oscillatory motion, we find that the frequency goes through an abrupt change for RS > 85,000, indicating two different modes of oscillation. When there is a gravity modulation at 1Hz with amplitude of 0.4G, no significant effect is found.

3 Numerical Simulation

A finite difference method algorithm for the simulation of the double diffusive convection in a rectangular domain has been implemented. Vorticity-streamfunction formulation is used for the fluid flow while the energy and concentration equations are written in the usual form. The various boundary and initial conditions are implemented. Dufort Frankel method is used to approximate the diffusive terms, Arakawa's 9-point scheme is used for the Jacobian terms, and central difference is used for the time derivative and buoyancy term. The Poisson equation (streamfunction) is solved by fast Fourier transform and cyclic reduction. The algorithm has been tested against previous known numerical results, driven square cavity and buoyancy convection. We used the algorithm to simulate convection similar to the experiment. It was found that the same type of convection is found. In fact, many of the salient features observed in the experiments are also obtained in the numerical simulations.

We use the developed algorithm to simulate the experimental conditions. The computational domain is 1 (height) by 11 (width). The dimensionless parameters are set to Pr = 7, Le = 118.3, and RT = 162,000. The initial conditions are and . We use the experimental step change concentration distribution and calculate the 1-D diffusion equation. The concentration distribution at 30 minutes is used as the initial condition for the numerical simulation of the double diffusive convection. To simulate the perturbation in the experimental setup, we introduce random disturbance of order 10E-6 to streamfunction, temperature, and concentration continuously.

For Steady G, the onset RS is found to be 124,000, which compares very well with the experimental result. The convection pattern is similar to the experimental observation, two-layer structure at onset and evolves to single layer structure. With gravity modulation, the critical RS was found to be 126,000, and frequency = 0.051Hz. These values differ from the experimental values by 7% and 10.5% respectively.

In an attempt to explain the difference between experiment and numerical simulation, we postulate that the variable viscosity is the key factor. Consequently, we extend our algorithm to incorporate variable viscosity effect. The results indicate that the variable viscosity is negligible.

4. Summary

	We have performed experiments to determine the critical conditions for onset of instabilities in a horizontal layer under steady and oscillating G and compared with the predictions of linear stability theory and the results of nonlinear simulation. The linear stability results for steady G agrees very well with the experiment but it fails to predict any effects of gravity oscillation. The nonlinear simulation gives results that are within 10% of the experimental value for both steady and oscillating G with the steady G results showing a better agreement than the oscillating G results.
Bibliography Type:	Description: (Last Updated: 09/11/2009)
Articles in Peer-reviewed Journals	Chan CL, Yu Y, Chen CF. "Instability of convection of an ethanol–water solution in a vertical tank." Journal of Fluid Mechanics, 2004 Jul;510:243-65. http://dx.doi.org/10.1017/S002211200400953X , Jul-2004
Papers from Meeting Proceedings	Chen CF, Chan CL, Yu Y. "Onset of double-diffusive instability under steady and sinusoidally varying gravity." 57th Annual Meeting of the Division of Fluid Dynamics, Seattle, WA, November 2004. Bulletin of the American Physical Society, 2004 Nov;49(9):65-6. , Nov-2004
Presentation	Chan, Cho Lik "Theoretical, experimental, and numerical modeling of natural convection" Department of Atmospheric Sciences, University of Arizona Feb-2004